STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-HOMOTOPY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3251-3256 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 3258-3296, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_algebraic-stacks-sheaves.json (pb=1577, stmt 3251-3256, proof 3258-3296).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0040: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3296–3296. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-generalities: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3212–3232. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *If there exists a 1-morphism $s : \mathcal{X} \to \mathcal{U}$ such that $f \circ s$ is 2-isomorphic to $id_{\mathcal{X}}$ then the extended Čech complex is homotopic to zero.*

Proof. Set $\mathcal{U}' = \mathcal{U} \times_{\mathcal{X}} \mathcal{X}$ equal to the fibre product as described in Categories, Lemma 0040. Set $f' : \mathcal{U}' \to \mathcal{X}$ equal to the second projection. Then $\mathcal{U} \to \mathcal{U}'$, $u \mapsto (u, f(x), 1)$ is an equivalence over $\mathcal{X}$, hence we may replace $(\mathcal{U}, f)$ by $(\mathcal{U}', f')$ by Lemma lemma-generalities. The advantage of this is that now f' has a section s' such that $f' \circ s' = \mathrm{id}_{\mathcal{X}}$ on the nose. Namely, if $t : s \circ f \to \mathrm{id}_{\mathcal{X}}$ is a 2-isomorphism then we can set $s'(x) = (s(x), x, t_x)$. Thus we may assume that $f \circ s = \mathrm{id}_{\mathcal{X}}$.

In the case that $f \circ s = \mathrm{id}_{\mathcal{X}}$ the result follows from general principles. We give the homotopy explicitly. Namely, for $n \geq 0$ define $s_n : \mathcal{U}_n \to \mathcal{U}_{n+1}$ to be the 1-morphism defined by the rule on objects

$$(u_0, \ldots, u_n, x, \alpha_0, \ldots, \alpha_n) \longmapsto (u_0, \ldots, u_n, s(x), x, \alpha_0, \ldots, \alpha_n, \mathrm{id}_x).$$

Define

$$h^{n+1} : \Gamma(\mathcal{U}_{n+1}, f_{n+1}^{-1}\mathcal{F}) \longrightarrow \Gamma(\mathcal{U}_n, f_n^{-1}\mathcal{F})$$

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as pullback along s_n. We also set $s_{-1} = s$ and $h^0 : \Gamma(\mathcal{U}_0, f_0^{-1}\mathcal{F}) \to \Gamma(\mathcal{X}, \mathcal{F})$ equal to pullback along s_{-1}. Then the family of maps $\{h^n\}_{n\geq 0}$ is a homotopy between 1 and 0 on the extended Čech complex. $\square$